

03/08/26

Signal Processing

- Recap: Discrete time signals and systems
- Frequency domain representation of signals and systems
 - Discrete Fourier Transform (DFT)
 - Discrete Time Fourier Transform (DTFT)
 - Examples
- * Aside: Look up Haar wavelets (Not going to be on the test)

- The notion of discrete frequency

- Two representative signals

- $x_1[n] = 1, 0 \leq n \leq 7,$
0, else.

- $x_2[n] = \cos \pi n, 0 \leq n \leq 7,$
0, else

- The Discrete Fourier Transform (DFT):

$$\begin{array}{ccc} X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi \frac{nk}{N}} & \xrightarrow{\text{signal length}} & \\ \downarrow & & \downarrow \\ \text{Discrete} & & \text{discrete} \\ \text{frequency} & & \text{time} \\ \text{index} & & \text{index} \end{array}$$

- Let's compute the DFT of $x_1[n]$ and $x_2[n]$

- $x_2[n] = \cos \pi n = \cos \left(\frac{2\pi}{2} n \right) \quad \frac{2\pi k}{N} = \frac{2\pi}{2}$